# U.S. Treasury Yield-Curve Factor Research

Historical factor decomposition, calendar regime comparisons and transaction-cost sensitivity

**RESEARCH ONLY:** Fed fitted-yield observations are generally published weekly and may be revised. These results do not reconstruct the information available on historical trading dates. They are not executable returns.

## Executive findings

Complete records: 9,189, 1990-01-02 to 2026-10-23. PC1: 83.9% of daily-change variance; PCs 1–3 jointly: 97.9%.

Model observations: 8,934 after warm-up; active modeled dates: 3,952; entry events: 94. All sensitivity units are hypothetical dollars per assumed $1/bp gross DV01.

## Observed fitted yields and factor loadings

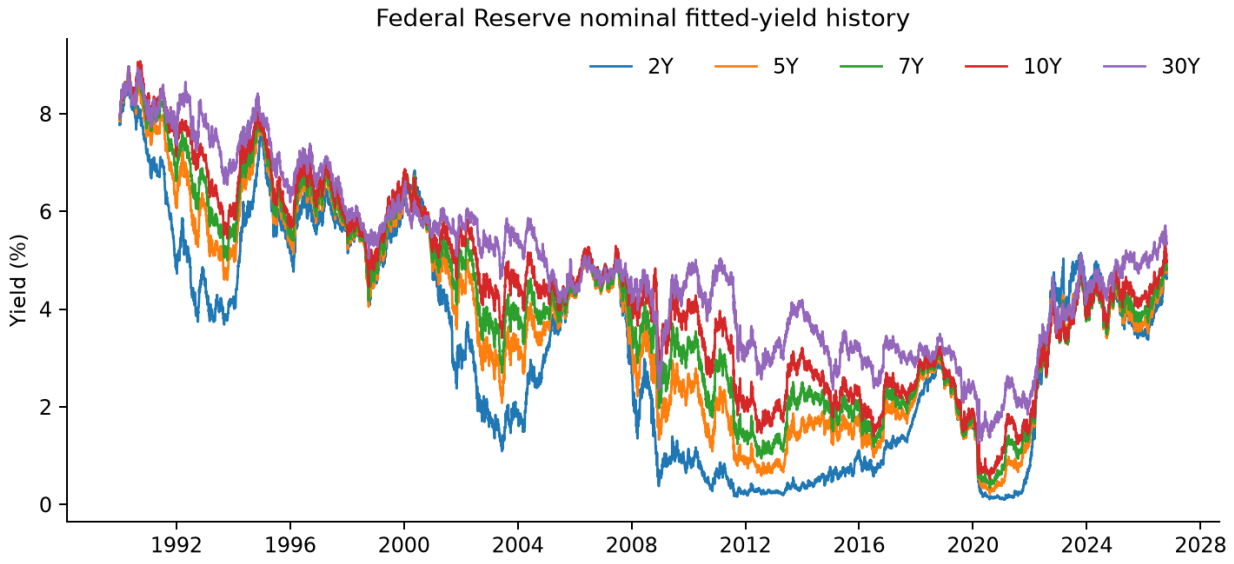



Figure 1. Model-derived nominal fitted yields, not cash-bond executable quotes.

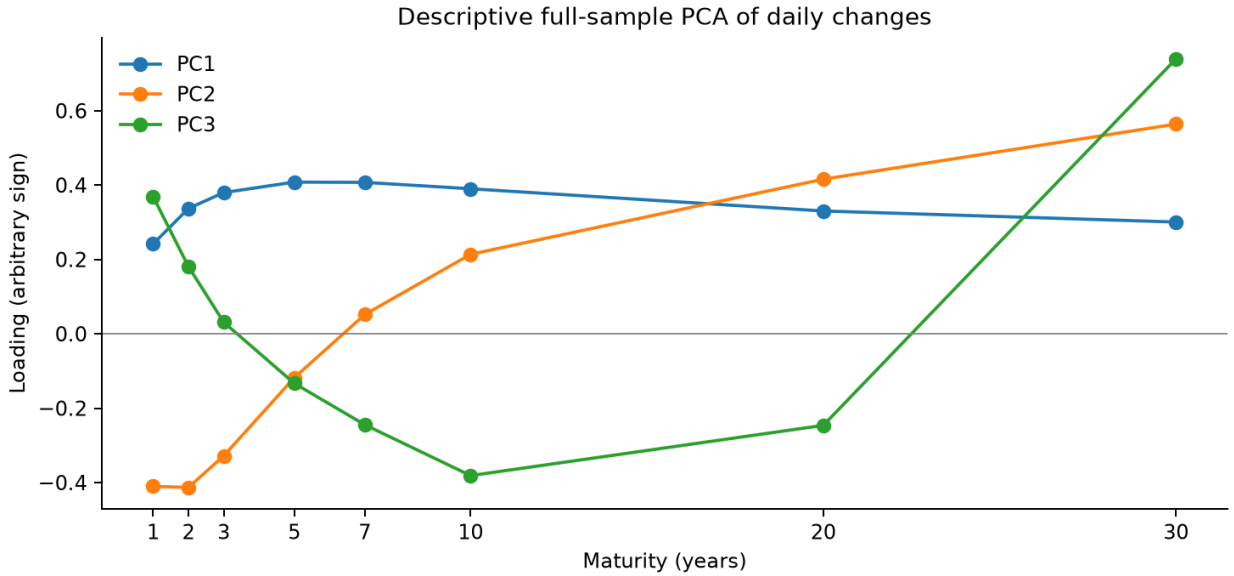



Figure 2. Descriptive PCA loadings; mathematical signs are arbitrary.

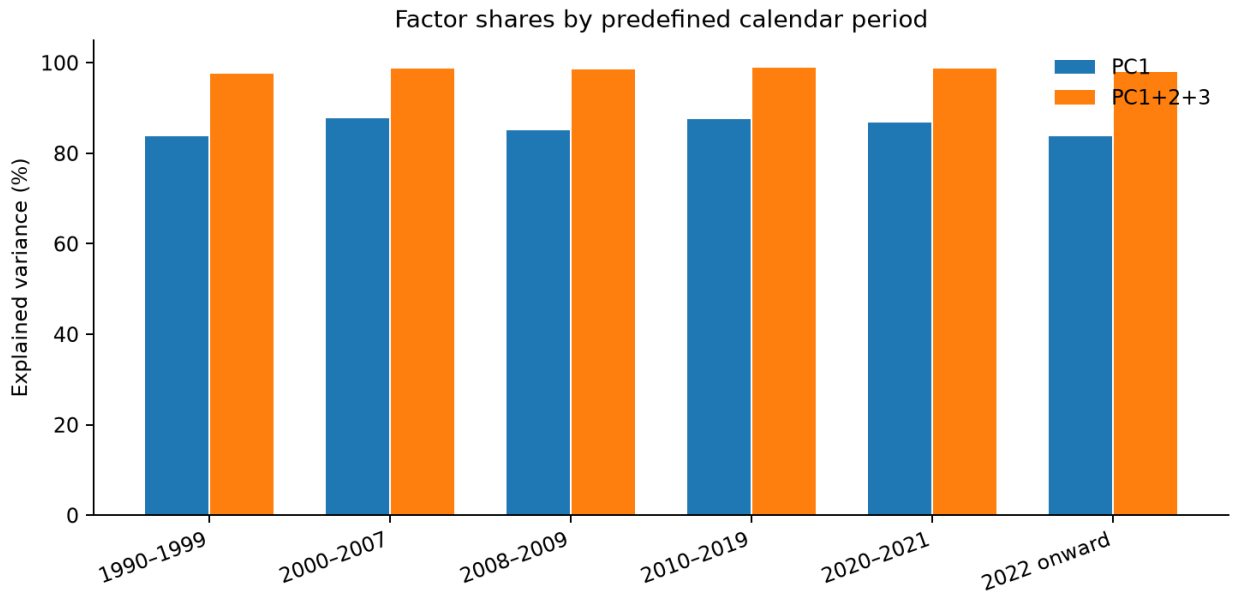



Figure 3. Variance share by predefined historical interval.

## Hypothetical sensitivity and assumed costs

A separate rolling 5Y–7Y–10Y minimum-variance fly model uses a 252-row PCA window, 126-row z-history, entry threshold 2.0 and exit threshold 0.5. Signals are shifted in observation rows; actual public-release availability is not modeled.

| Assumed cost (bp) | Gross $ | Cost $ | Net $ |
|---|---|---|---|
| 0.00 | 4.74 | 0.00 | 4.74 |
| 0.25 | 4.74 | 48.26 | -43.52 |
| 0.50 | 4.74 | 96.53 | -91.78 |
| 1.00 | 4.74 | 193.05 | -188.31 |
| 2.00 | 4.74 | 386.10 | -381.36 |

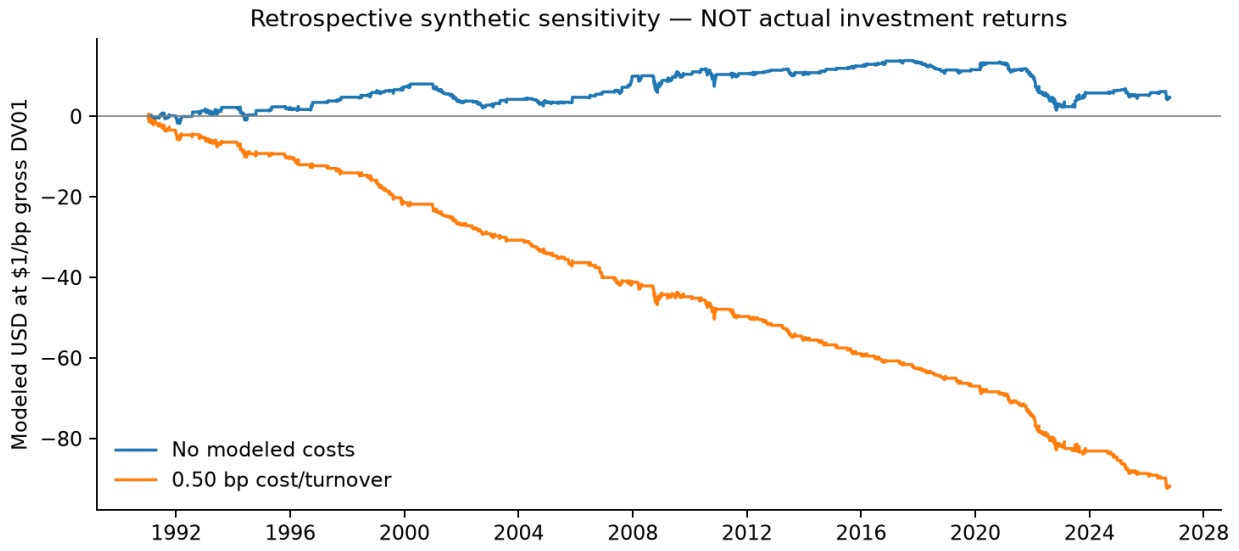



Figure 4. Retrospective synthetic cumulative sensitivity; NOT a real-bond P&L; series.

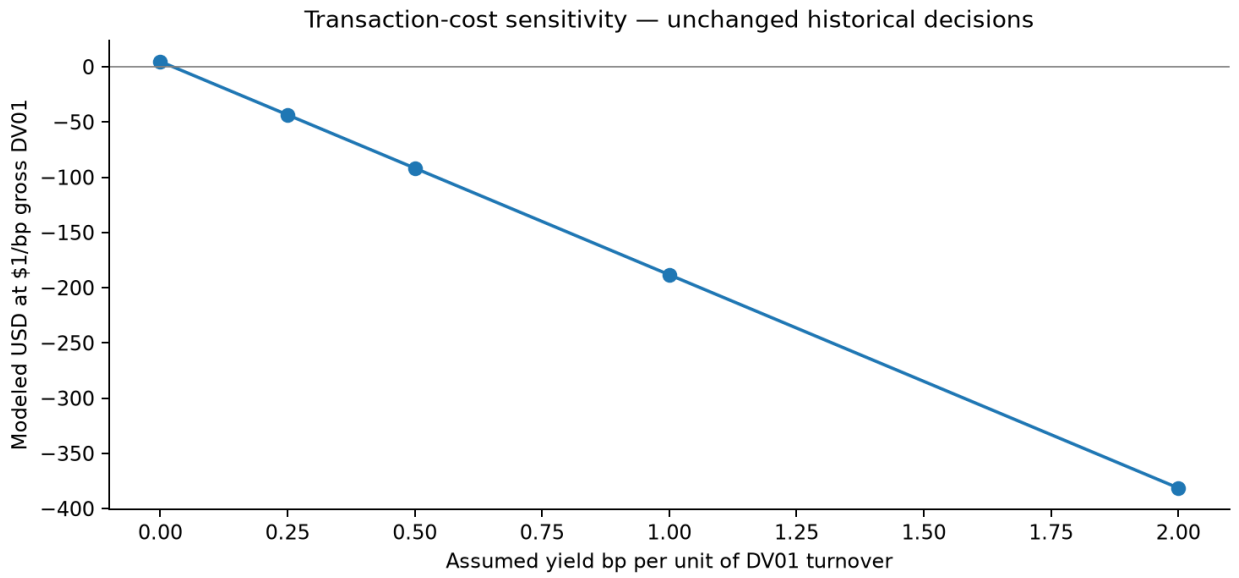



Figure 5. Net sensitivity under different hypothetical unit-turnover costs.

## Limitations and sources

Data are current-vintage Fed fitted nominal yields. No historical release timestamps, live bond CUSIPs, coupon carry, financing, executable bid/ask, or investment capital are modeled. Cross-era summaries are descriptive, not out-of-sample validation.

CSV SHA-256: 88cab90e84ad765ab79d9854b33e5813051210f2ac2285eaa6e720149fdd256d

Source: https://www.federalreserve.gov/data/nominal-yield-curve.htm